Teaching the robotics club

A practical workshop for high school mentors

MechaRAMS · Proposed workshop time: 120 minutes

The robot setup

A Left wheel motor B Right wheel motor

C Forward distance sensor

D Downward color sensor

The auxiliary attachment motor stays unplugged.

A recipe inside a complete program

```
travel = 180
await drive(travel)
await signal("D")
```

Students edit the recipe. Mentors prepare the support code.

Sensor decisions

A valid gap of 150 mm or less means parking is close.

A negative distance reading means check the sensor.

A dark marker needs a measured reflection threshold.

A movement limit means the target was not detected.

A small attachment and a fair comparison

Build and hand-test a guide for a lightweight cube.

Keep wheels and the color sensor clear.

Test two designs with the same program and start.

Store each team’s removable attachment separately.

The coaching conversation

What did you expect?

What did you observe?

What one change could test your idea?

Two teams of three share each robot

14–26 minutes T1/T3 test. T2/T4 build.

26–38 minutes Teams switch stations.

38–50 minutes Each team gets a six-minute retest.

Rotate three jobs in equal thirds of each turn.

Ready to lead a pod

Connect, stop, and restore a known-good project.

Explain this week’s idea and its common mistakes.

Give all three students meaningful operating turns.

Save the team’s work and record what actually happened.